

Answer Key — Biology Grade 11

Sub-Strand 3.2: Growth and Development in Animals • Lesson 6: Final Explanation: Controlled Growth, Metamorphosis and Survival in Kenyan Insects

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	B	D	D	A	C	A	B

USE AFTER: OBSERVE PHASE — Observe: growth curve graphs from Lesson 3 showing intermittent (step-like) growth in insect body length

Q1. A graph shows the body length of a grasshopper nymph over time. The line rises sharply at some points and stays flat between them. What best explains the flat sections?

- A. The nymph stops eating and starts to shrink
- B. The nymph has become an adult and has stopped moulting
- C. The hard exoskeleton cannot stretch, so the nymph cannot grow bigger until it moults**
- D. Hormones are completely absent from the body during these times

Correct answer: C. The exoskeleton is rigid and cannot stretch, so length stays constant between moults. Growth happens in a burst just after moulting, while the new exoskeleton is still soft.

USE AFTER: OBSERVE PHASE — Observe: recalculating one growth rate from the graph for a step phase using the Lesson 3 formula

Q2. During one step (moult) phase, a grasshopper nymph grows from 14 mm on Day 10 to 20 mm on Day 13. Using growth rate = change in length ÷ time taken, what is its growth rate in this phase?

- A. 6 mm per day
- B. 2 mm per day**
- C. 6.7 mm per day
- D. 0.5 mm per day

Correct answer: B. Change in length = $20 - 14 = 6$ mm. Time taken = $13 - 10 = 3$ days. Growth rate = $6 \div 3 = 2$ mm per day.

USE AFTER: OBSERVE PHASE — Observe: hormone summary (ecdysone triggers moulting, juvenile hormone keeps the juvenile form)

Q3. In a caterpillar, which statement about ecdysone and juvenile hormone is correct?

- A. Juvenile hormone triggers moulting, and ecdysone keeps the insect in the larval form
- B. Both hormones stop the caterpillar from moulting
- C. Ecdysone makes the exoskeleton stretch so no moulting is needed
- D. Ecdysone triggers moulting, and a high level of juvenile hormone makes the moult produce another larval stage**

Correct answer: D. Ecdysone triggers moulting. When juvenile hormone is high, the moult gives another larva or nymph. When it is low, pupation or adult formation can happen.

USE AFTER: EXPLAIN PHASE — Explain: CER poster stage-by-stage contrast between complete and incomplete metamorphosis

Q4. Which pair correctly matches each insect with its life-cycle stages?

- A. Butterfly: egg, larva, pupa, adult. Grasshopper: egg, larva, pupa, adult
- B. Butterfly: egg, nymph, adult. Grasshopper: egg, nymph, adult
- C. Butterfly: egg, nymph, adult. Grasshopper: egg, larva, pupa, adult
- D. Butterfly: egg, larva, pupa, adult. Grasshopper: egg, nymph, adult**

Correct answer: D. A butterfly has complete metamorphosis (egg, larva, pupa, adult). A grasshopper has incomplete metamorphosis (egg, nymph, adult), and its nymph looks like a small adult.

USE AFTER: EXPLAIN PHASE — Explain: CER reasoning on optimising resource use and reducing parent-offspring competition (caterpillar on sukuma wiki, butterfly drinking nectar)

Q5. A caterpillar eats sukuma wiki leaves, while the adult butterfly drinks nectar from flowers. How does this difference help the species survive?

- A. The larva and the adult do not compete for the same food**
- B. The caterpillar can use nectar when leaves run out
- C. It lets the butterfly avoid growing and moulting
- D. The butterfly no longer needs any food after the pupa stage

Correct answer: A. In complete metamorphosis the larva and the adult use different foods. This reduces competition between young and adults, and so uses resources efficiently.

USE AFTER: EXPLAIN PHASE — Explain: Gallery Walk peer rubric check that the old exoskeleton is replaced, not peeled like skin

Q6. A student finds an empty, split insect skin on a maize leaf beside a larger grasshopper. Which statement is the best scientific explanation?

- A. The grasshopper peeled its skin like sunburned skin because it was damaged by the sun
- B. The grasshopper melted inside its skin and rebuilt a new body without any control
- C. The grasshopper shed its old rigid exoskeleton during moulting, triggered by ecdysone, and a new larger exoskeleton replaced it**
- D. The skin split because the grasshopper stretched it smoothly as it grew

Correct answer: C. The exoskeleton cannot stretch. Ecdysone triggers moulting, the old exoskeleton splits and is shed, and a new, larger one hardens in its place.

USE AFTER: DQB PHASE — DQB: sorting sticky-note questions into 'Answered (with evidence)', 'Partly answered' and 'New questions'

Q7. During the Driving Question Board sorting, which question can the class place under 'Answered (with evidence)' using evidence from this unit?

- A. Why does the growth curve of an insect have steps instead of a smooth line?**
- B. How might human pesticides disrupt hormonal pathways in agricultural pests?
- C. Do other animals, such as crabs in Mombasa, moult like insects?
- D. Which new chemicals could block ecdysone in crop pests?

Correct answer: A. The step-like growth curve was explained with evidence from exuviae, the rigid exoskeleton, moulting and graphs. The other questions go beyond the unit and are new questions.

USE AFTER: MODEL BUILDING PHASE — Model: Final Model showing the intermittent growth curve and Day 1 vs Final comparison table

Q8. In their Day 1 versus Final comparison, a learner writes about why a calf and a grasshopper nymph grow differently. Which statement is correct?

- A. Neither animal needs hormones to control growth
- B. A calf grows in a sigmoid (S-shaped) pattern without an exoskeleton limit, while an insect grows in steps because its exoskeleton must be replaced**
- C. A calf grows in steps and an insect grows smoothly because insects are smaller
- D. Both grow in steps because both have a hard exoskeleton

Correct answer: B. A calf has no rigid outer skeleton, so it grows continuously, giving a sigmoid curve. An insect can only increase in size after a moult, giving intermittent, step-like growth.